```c
// C Program: Factorial and Fibonacci Series

#include <stdio.h>

int main()
{
    int n, i;
    long long fact = 1;
    int a = 0, b = 1, c;

    printf("Enter a non-negative integer: ");
    scanf("%d", &n);

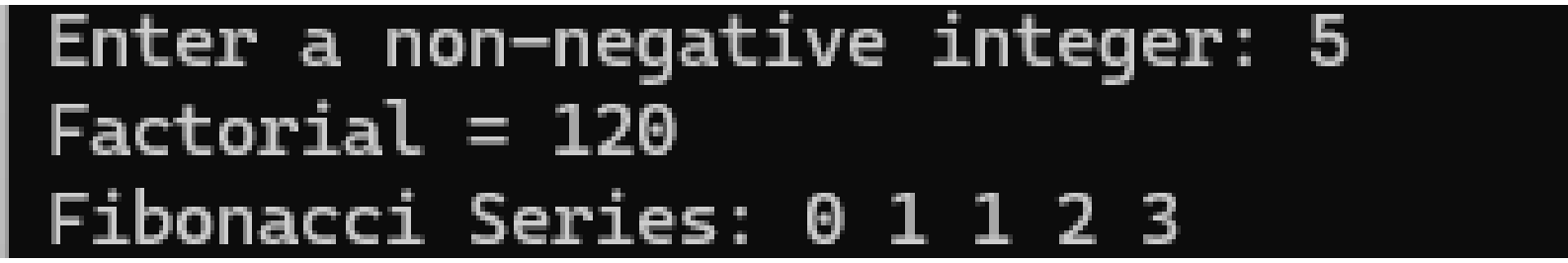
    // Factorial
    for(i = 1; i <= n; i++)
        fact = fact * i;

    printf("Factorial = %lld\n", fact);

    // Fibonacci series
    printf("Fibonacci Series: ");
    for(i = 1; i <= n; i++)
    {
        printf("%d ", a);
        c = a + b;
```

```
        a = b;
        b = c;
    }

    return 0;
}
```


Enter a non-negative integer: 5
Factorial = 120
Fibonacci Series: 0 1 1 2 3

12 Right-Angled Number Pattern

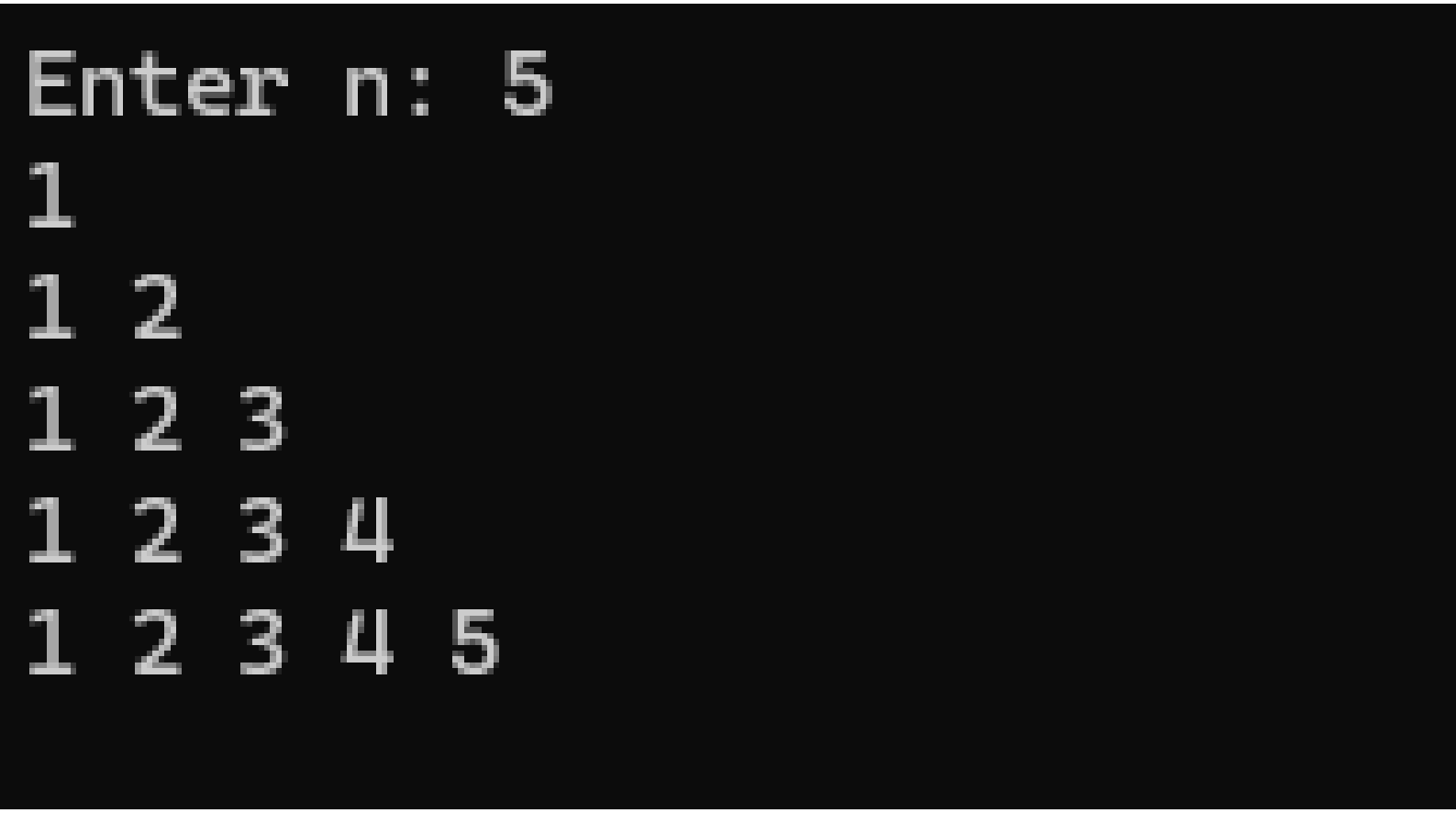
```c
#include <stdio.h>

int main() {
    int n, i, j;

    printf("Enter n: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        for(j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }

    return 0;
}
```


Enter n: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

13. Array Operations

```c
#include <stdio.h>

int main()
{
    int a[100], n, i;
    int sum = 0, max, min;
    int even = 0, odd = 0;
    float average;

    printf("Enter n: ");
    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    max = min = a[0];

    for(i = 0; i < n; i++)
    {
        sum = sum + a[i];

        if(a[i] > max)
            max = a[i];

        if(a[i] < min)
            min = a[i];

        if(a[i] % 2 == 0)
            even++;
        else
            odd++;
    }
```

```c
    average = (float)sum / n;

    printf("Sum = %d\n", sum);
    printf("Average = %.2f\n", average);
    printf("Largest = %d\n", max);
    printf("Smallest = %d\n", min);
    printf("Even elements = %d\n", even);
    printf("Odd elements = %d\n", odd);

    return 0;
}
```

```
Enter n: 5
Enter elements:
10 4 8 3 2
Sum = 27
Average = 5.40
Largest = 10
Smallest = 2
Even elements = 4
Odd elements = 1
```

14. Linear Search

```c
#include <stdio.h>

int main()
{
    int a[100], n, key, i, count = 0;

    printf("Enter n: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    for(i = 0; i < n; i++)
    {
        if(a[i] == key)
        {
            printf("Position = %d\n", i + 1);
            count++;
        }
    }

    if(count == 0)
        printf("Element not found\n");
    else
        printf("Frequency = %d\n", count);

    return 0;
}
```

```
Enter n: 5
Enter elements:
10 20 10 30 10
Enter element to search: 10
Position = 1
Position = 3
Position = 5
Frequency = 3
```

15. Bubble Sort - Ascending and Descending

```c
#include <stdio.h>

int main()
{
    int a[100], n;
    int i, j, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Original Array: ");

    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }

    /* Ascending */
    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
```

```c
/* Ascending */
for(i = 0; i < n - 1; i++)
{
    for(j = 0; j < n - i - 1; j++)
    {
        if(a[j] > a[j + 1])
        {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
        }
    }
}

printf("\nAscending Order: ");

for(i = 0; i < n; i++)
{
    printf("%d ", a[i]);
}

/* Descending */
for(i = 0; i < n - 1; i++)
{
    for(j = 0; j < n - i - 1; j++)
    {
        if(a[j] < a[j + 1])
        {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
        }
    }
}
```

```
    printf("\nDescending Order: ");

    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

```
Enter number of elements: 5
Enter elements:
40 10 50 20 30
Original Array: 40 10 50 20 30
Ascending Order: 10 20 30 40 50
Descending Order: 50 40 30 20 10
```

16. Matrix Addition, Subtraction, Transpose and Multiplication

```c
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r, col, i, j;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &col);

    printf("Enter Matrix A:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter Matrix B:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    /* Addition */
    printf("\nAddition:\n");

    for(i = 0; i < r; i++)
    {
```

```
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            c[i][j] = a[i][j] + b[i][j];
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }

    /* Subtraction */
    printf("\nSubtraction:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            c[i][j] = a[i][j] - b[i][j];
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }

    /* Transpose */
    printf("\nTranspose of A:\n");

    for(i = 0; i < col; i++)
    {
        for(j = 0; j < r; j++)
        {
            printf("%d ", a[j][i]);
        }
        printf("\n");
    }

    return 0;
}
```

```
Enter rows and columns: 2 2
Enter Matrix A:
1 2 3 4
Enter Matrix B:
5 6 7 8

Addition:
6 8
10 12

Subtraction:
-4 -4
-4 -4

Transpose of A:
1 3
2 4
```

17. String Operations Without Predefined Functions

```c
#include <stdio.h>

int main()
{
    char str1[50], str2[50];
    int i, j, len = 0;
    int same = 1;

    printf("Enter a string: ");
    scanf("%s", str1);

    /* Length */
    for(i = 0; str1[i] != '\0'; i++)
    {
        len++;
    }

    printf("Length = %d\n", len);

    /* Copy */
    for(i = 0; str1[i] != '\0'; i++)
    {
        str2[i] = str1[i];
    }

    str2[i] = '\0';

    printf("Copied string = %s\n", str2);

    /* Reverse */
    printf("Reverse = ");

    for(i = len - 1; i >= 0; i--)
```

```c
    for(i = len - 1; i >= 0; i--)
    {
        printf("%c", strl[i]);
    }

    /* Palindrome */
    for(i = 0; i < len / 2; i++)
    {
        if(strl[i] != strl[len - 1 - i])
        {
            same = 0;
            break;
        }
    }

    if(same == 1)
        printf("\nPalindrome");
    else
        printf("\nNot Palindrome");

    return 0;
}
```

```
Enter a string: keerthi
Length = 7
Copied string = keerthi
Reverse = ihtreek
Not Palindrome
```

```
Enter a string: madam
Length = 5
Copied string = madam
Reverse = madam
Palindrome
```

18. GCD, LCM, Factorial and Power Using Functions/Recursion

```c
#include <stdio.h>

int gcd(int a, int b)
{
    if(b == 0)
        return a;

    return gcd(b, a % b);
}

int factorial(int n)
{
    if(n == 0)
        return 1;

    return n * factorial(n - 1);
}

int power(int x, int n)
{
    if(n == 0)
        return 1;

    return x * power(x, n - 1);
}

int lcm(int a, int b)
{
    return (a * b) / gcd(a, b);
}

int main()
{
    int choice, a, b, n, x;
```

```c
int choice, a, b, n, x;

printf("1. GCD\n");
printf("2. LCM\n");
printf("3. Factorial\n");
printf("4. Power\n");

printf("Enter choice: ");
scanf("%d", &choice);

if(choice == 1)
{
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("GCD = %d", gcd(a, b));
}
else if(choice == 2)
{
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("LCM = %d", lcm(a, b));
}
else if(choice == 3)
{
    printf("Enter number: ");
    scanf("%d", &n);

    printf("Factorial = %d", factorial(n));
}
else if(choice == 4)
{
    printf("Enter x and n: ");
    scanf("%d %d", &x, &n);

    printf("Answer = %d", power(x, n));
```

```
        printf("Enter x and n: ");
        scanf("%d %d", &x, &n);

        printf("Answer = %d", power(x, n));
    }
    else
    {
        printf("Invalid choice");
    }

    return 0;
}
```

```
1. GCD
2. LCM
3. Factorial
4. Power
Enter choice: 1
Enter two numbers: 12 18
GCD = 6
```

```
1. GCD
2. LCM
3. Factorial
4. Power
Enter choice: 2
Enter two numbers: 4 6
LCM = 12
```

```
1. GCD
2. LCM
3. Factorial
4. Power
Enter choice: 3
Enter number: 5
Factorial = 120
```

19. Dynamic Memory Allocation Using malloc()

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *p, n, i;
    int sum = 0;
    int largest, smallest;
    float average;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    p = (int *)malloc(n * sizeof(int));

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", p + i);
    }

    largest = *p;
    smallest = *p;

    for(i = 0; i < n; i++)
    {
        sum = sum + *(p + i);

        if(*(p + i) > largest)
            largest = *(p + i);

        if(*(p + i) < smallest)
            smallest = *(p + i);
    }
```

```
        if(*(p + i) < smallest)
            smallest = *(p + i);
    }

    average = (float)sum / n;

    printf("Sum = %d\n", sum);
    printf("Average = %.2f\n", average);
    printf("Largest = %d\n", largest);
    printf("Smallest = %d\n", smallest);

    free(p);

    return 0;
}
```

```
Enter number of elements: 5
Enter elements:
10 20 30 40 50
Sum = 150
Average = 30.00
Largest = 50
Smallest = 10
```

20 Student Record

```c
#include <stdio.h>

struct Student
{
    int regno;
    char name[30];
    int m1, m2, m3;
    int total;
    float percentage;
};

int main()
{
    struct Student s[50];
    int n = 0;
    int choice, i, search;
    FILE *fp;

    do
    {
        printf("\n\n--- STUDENT MENU ---");
        printf("\n1. Add Record");
        printf("\n2. Display Records");
        printf("\n3. Search Record");
        printf("\n4. Write to File");
        printf("\n5. Read from File");
        printf("\n6. Exit");

        printf("\nEnter choice: ");
        scanf("%d", &choice);
```

```c
/* 1. Add Record */
if(choice == 1)
{
    printf("\nEnter Register Number: ");
    scanf("%d", &s[n].regno);

    printf("Enter Name: ");
    scanf("%s", s[n].name);

    printf("Enter 3 Marks: ");
    scanf("%d %d %d", &s[n].m1, &s[n].m2, &s[n].m3);

    s[n].total = s[n].m1 + s[n].m2 + s[n].m3;

    s[n].percentage = s[n].total / 3.0;

    n++;

    printf("Record added successfully!");
}

/* 2. Display Records */
else if(choice == 2)
{
    for(i = 0; i < n; i++)
    {
        printf("\nRegister Number = %d", s[i].regno);
        printf("\nName = %s", s[i].name);
        printf("\nTotal = %d", s[i].total);
        printf("\nPercentage = %.2f", s[i].percentage);

        if(s[i].m1 >= 35 &&
           s[i].m2 >= 35 &&
           s[i].m3 >= 35)
        {
            printf("\nResult = Pass");
        }
```

```
            printf("\nResult = Pass");
        }
        else
        {
            printf("\nResult = Fail");
        }
    }
}

/* 3. Search Record */
else if(choice == 3)
{
    printf("\nEnter Register Number: ");
    scanf("%d", &search);

    for(i = 0; i < n; i++)
    {
        if(s[i].regno == search)
        {
            printf("\nRecord Found!");
            printf("\nName = %s", s[i].name);
            printf("\nTotal = %d", s[i].total);
            printf("\nPercentage = %.2f",
                   s[i].percentage);
            break;
        }
    }

    if(i == n)
    {
        printf("\nRecord Not Found!");
    }
}

/* 4. Write to File */
else if(choice == 4)
{
    fp = fopen("student.txt", "w");
```

```
    if(fp == NULL)
    {
        printf("File cannot be opened!");
    }
    else
    {
        for(i = 0; i < n; i++)
        {
            fprintf(fp, "%d %s %d %d %d %d %.2f\n",
                    s[i].regno,
                    s[i].name,
                    s[i].m1,
                    s[i].m2,
                    s[i].m3,
                    s[i].total,
                    s[i].percentage);
        }

        fclose(fp);

        printf("Records saved to file!");
    }
}

/* 5. Read from File */
else if(choice == 5)
{
    fp = fopen("student.txt", "r");

    if(fp == NULL)
    {
        printf("File not found!");
    }
    else
    {
        printf("\n--- Records From File ---\n");

        while(fscanf(fp, "%d %s %d %d %d %d %f",
```

```
            while(fscanf(fp, "%d %s %d %d %d %d %f",
                         &s[0].regno,
                         s[0].name,
                         &s[0].m1,
                         &s[0].m2,
                         &s[0].m3,
                         &s[0].total,
                         &s[0].percentage) == 7)
            {
                printf("\nRegister Number = %d",
                       s[0].regno);

                printf("\nName = %s", s[0].name);

                printf("\nTotal = %d", s[0].total);

                printf("\nPercentage = %.2f\n",
                       s[0].percentage);
            }

            fclose(fp);
        }
    }

    /* 6. Exit */
    else if(choice == 6)
    {
        printf("\nProgram Ended!");
    }

    else
    {
        printf("\nInvalid Choice!");
    }

} while(choice != 6);
```

```
        }

    } while(choice != 6);

    return 0;
}
```

```
--- STUDENT MENU ---
1. Add Record
2. Display Records
3. Search Record
4. Write to File
5. Read from File
6. Exit
Enter choice: 1

Enter Register Number: 101
Enter Name: keerthi
Enter 3 Marks: 80 75 90
Record added successfully!

--- STUDENT MENU ---
1. Add Record
2. Display Records
3. Search Record
4. Write to File
5. Read from File
6. Exit
Enter choice: 2

Register Number = 101
Name = keerthi
Total = 245
Percentage = 81.67
Result = Pass
```

```
--- STUDENT MENU ---
1. Add Record
2. Display Records
3. Search Record
4. Write to File
5. Read from File
6. Exit
Enter choice: 3

Enter Register Number: 101

Record Found!
Name = keerthi
Total = 245
Percentage = 81.67

--- STUDENT MENU ---
1. Add Record
2. Display Records
3. Search Record
4. Write to File
5. Read from File
6. Exit
Enter choice: 4
Records saved to file!
```

```
--- STUDENT MENU ---
1. Add Record
2. Display Records
3. Search Record
4. Write to File
5. Read from File
6. Exit
Enter choice: 5

--- Records From File ---

Register Number = 101
Name = keerthi
Total = 245
Percentage = 81.67


--- STUDENT MENU ---
1. Add Record
2. Display Records
3. Search Record
4. Write to File
5. Read from File
6. Exit
Enter choice: 6

Program Ended!
```